



IDENTIFICATION

CODE : GI-3-S2-EC-CBD
ECTS : 3

HOURS

Cours : 0h
TD : 30h
TP : 16h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 48h
Travail personnel : 8h
Total : 56h

ASSESSMENT METHOD

IE (interrogation écrite)
ES (évaluation en situation)

TEACHING AIDS

Slides

TEACHING LANGUAGE

French

CONTACT

Mme SANDOZ-GUERMOND
Françoise :
francoise.sandoz-guermond@insa-lyon.fr

AIMS

This course belongs to teaching unit Conception et gestion des systèmes d'informations (GI-3-S2-UE-CGSI) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 2)

A2 Operate a model of a real or virtual system (Level 2)

A5 Process data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- (A1, C2)

To be able to :

- (A1, A5, C2, A2)

* Design and size systems required for a given activity of production, service, distribution

...

* Identify and correct discrepancies in a continuous improvement process (material, human and information systems) by the implementation of analytical and simulation tools

* Ensure quality reporting through the establishment of appropriate indicators, based on a systemic view of organizations (matrix, silo, project) for any scope and any type of activity"

CONTENT

Know :

- Architecture of a Data base system management

- Model entity - association

- Fonctionnal and multivaluate dependances

- Transitive closure notion, Minimal cover

- Normal forms

- Relationnal algebra, calculus

- SQL language

- Structure of a XML document, Wellformed XML document

- Document type definition, XSLT document

To be able to :

- Design an entity/association model

- Translate an entity/association model to relationnal model

- Write queries in relationnel algebra, in SQL

- Write a well formed XML document

- Validate a XML document from a DTD

- Transform a XML document with XSLT language

BIBLIOGRAPHY

C. Chrismet, "Mise en oeuvre des bases de données" ; Ed. Eyrolles 1990

H.F. Korth, A. Silbershats, "Systèmes de gestion de base de données", Mc Graw-Hill, 1988

C. Marée, G. Ledant, "SQL : Initiation, programmation et maîtrise" - Armand Colin, 1994

K. Williams, M. Brundage : XML et les bases de données - Ed. Eyrolles 2001

PRE-REQUISITES

